

Experiment 9 – Automobile Speed Control using ANFIS Hybrid System

1. Theory

- **ANFIS (Adaptive Neuro-Fuzzy Inference System)** merges fuzzy logic and neural networks.
 - Fuzzy rules express human-like reasoning (“If slope $\uparrow$ and speed $\downarrow \rightarrow$ increase throttle”).
 - Neural learning adjusts membership functions and rule weights automatically.
 - The system learns input–output relations from data to maintain vehicle speed under varying road slopes and loads.
 - Uses a **Sugeno-type FIS** trained by a **hybrid algorithm** (least-squares + back-propagation).
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2. Working (Detailed)

1. Inputs (Slope Angle, Throttle %)

- Represent external driving conditions and driver’s torque demand.
- Provide real-time data that influence vehicle acceleration and control response.

2. Error (e) and Change in Error (de)

- Error = (Desired Speed – Actual Speed); Change in Error shows how fast the error is varying.
- These values form the control feedback that guides throttle adjustments.

3. Scaling with Gains (Ke, Kde, Ku)

- Ke and Kde normalize the input signals to fit within fuzzy inference limits.
- Ku scales the final output signal to suitable actuator range for throttle/brake control.

4. ANFIS Controller Core (Fuzzy + Neural Integration)

- Fuzzy logic layer interprets linguistic rules such as “If error is small and decreasing $\rightarrow$ reduce throttle.”
- Neural learning layer continuously tunes membership functions and rule weights for optimal behavior.

5. ACO Optimization Loop (Metaheuristic Layer)

- Ant Colony Optimization fine-tunes Ke, Kde, Ku, λ , μ to minimize control error metrics (IAE, ISE, RMSE).

- Ants explore multiple parameter paths and reinforce best-performing combinations through pheromone updating.

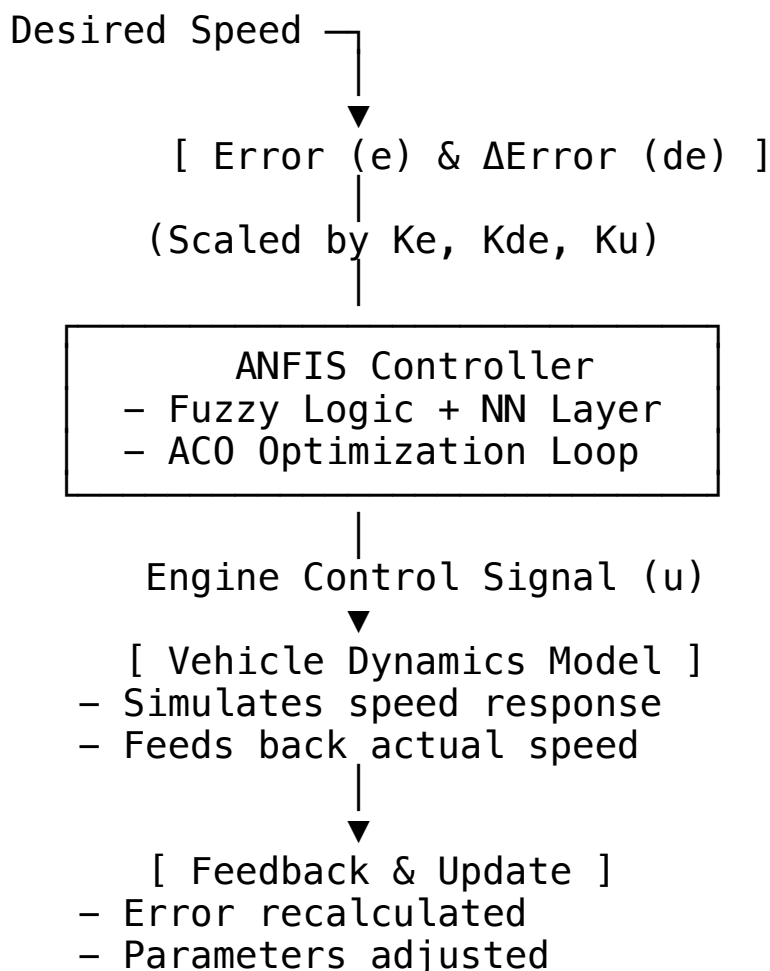
6. Vehicle Dynamics Block (Plant Model)

- Receives control signal $u(t)$ and converts it into vehicle speed response based on inertia and load.
- Provides nonlinear simulation of real-world conditions like slope variation and friction.

7. Feedback and Convergence

- The actual speed is continuously fed back to recalculate the new error.
- Once minimal error and steady-state stability are achieved, final optimal parameters are fixed for the ANFIS controller.

3. Block Diagram (with Explanation)



4. Advantages

- Automatically adapts to nonlinear and varying road conditions.
- Requires little manual tuning once trained.
- Provides smooth and stable speed response.

- Combines interpretability of fuzzy logic with learning power of neural nets.
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5. Disadvantages

- High computational cost during training.
 - Performance depends on quality of training data.
 - Choosing initial membership functions is critical.
 - Real-time hardware implementation may be resource-intensive.
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